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10 CONCEPTUAL Q&A CLOUD · PRACTICE WORKSHEET

CONCEPTUAL SERIES

BigQuery

Name: ______ Cohort/Batch: ______ Date: ______ Score: ______

BIGQUERY PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Cloud

TOTAL: 10 QUESTIONS

Q1 What is BigQuery and what is its primary purpose in its cloud ecosystem? *Threat Model*

Q2 How does IAM work with BigQuery? *Architecture*

Q3 How do you monitor BigQuery in production? *Configuration*

Q4 How do you scale BigQuery to handle variable load? *Hardening*

Q5 How do you secure BigQuery in production? *SSO / IdP*

Q6 How do you manage configuration and secrets for BigQuery? *Observability*

Q7 What are the main cost drivers for BigQuery and how do you optimize them? *Lifecycle*

Q8 How do you implement high availability with BigQuery? *Compliance*

Q9 How does BigQuery integrate with a CI/CD pipeline? *Testing*

Q10 What are common pitfalls when first adopting BigQuery? *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 BigQuery is a managed cloud service

Mitigation

BigQuery is a managed cloud service that handles a specific infrastructure concern (compute, storage, messaging, AI). Using it shifts operational burden to the cloud provider while you focus on application logic.

A2 Access to BigQuery is controlled via

Primitives

Access to BigQuery is controlled via IAM roles and policies. Attach the minimum required permissions to a service account or role. Avoid long-lived access keys; use instance roles or Workload Identity Federation instead.

A3 BigQuery emits metrics to the cloud

Hardening

BigQuery emits metrics to the cloud provider's monitoring service (CloudWatch, Cloud Monitoring, Azure Monitor). Set alerts on key metrics (error rate, latency, capacity). Use distributed tracing to correlate across services.

A4 BigQuery supports auto-scaling policies based on

Gotchas

BigQuery supports auto-scaling policies based on CPU, queue depth, or custom metrics. Set minimum and maximum capacity. Horizontal scaling (more instances) is generally preferred over vertical for cloud workloads.

A5 Place BigQuery in a private subnet

Federation

Place BigQuery in a private subnet with no public endpoint where possible. Use VPC security groups or firewall rules to allow only required traffic. Encrypt data at rest and in transit. Rotate credentials regularly.

A6 Store sensitive values in a managed

Observability

Store sensitive values in a managed secret store (AWS Secrets Manager, GCP Secret Manager, Azure Key Vault). Inject secrets as environment variables at runtime, never bake them into container images or source code.

A7 Cost in BigQuery typically comes from

Automation

Cost in BigQuery typically comes from compute hours, data transfer, storage, and request counts. Right-size instances, use reserved capacity for steady-state workloads, and enable lifecycle policies to expire old data.

A8 Deploy BigQuery across multiple availability

Governance

Deploy BigQuery across multiple availability zones. Use managed replicas or active-active configurations. Implement health checks and automatic failover. Test resilience regularly with chaos engineering or failover drills.

A9 CI/CD pipelines use the cloud CLI

Assessment

CI/CD pipelines use the cloud CLI or SDK to deploy updates to BigQuery after tests pass. Infrastructure-as-code (Terraform, CDK) defines BigQuery configuration as version-controlled code deployed via the pipeline.

A10 Common mistakes include misconfigured IAM

Optimization

Common mistakes include misconfigured IAM policies (too permissive), no cost alerting, deploying only in one availability zone, and missing backups. Read the service SLA carefully to understand what the cloud provider guarantees.